Manuel Rodríguez Villegas

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Profile

AI engineer with a strong mathematical background, building systems that see, learn and act: drones that fly from camera pixels, mapping robots and AI agents that assist engineers on their everyday work.

Education

Comillas Pontifical University, ETSI ICAI — Madrid, Spain
Master's Degree in Artificial Intelligence; GPA: 9.08/10; Class Representative — *2025 – 2026*
- Honors in Information Geometry.

Cornell University — Ithaca, NY, USA
Exchange Student - Electrical & Computer Engineering — *Jan – May 2025*

Comillas Pontifical University, ETSI ICAI — Madrid, Spain
Bachelor's Degree in Mathematical Engineering and Artificial Intelligence; GPA: 8.75/10 — *2021 – 2025*
- Honors in Probability & Statistics, Dynamic Systems, Big Data Architectures, Differential Geometry, Cybersecurity, Advanced Mathematics and Computer Vision II.

Experience

AUDI AG — Ingolstadt, Germany
AI Engineering Intern — *Jul – Nov 2026*
- Building an internal AI coworker on Azure for requirements engineers: it queries a graph database of 500,000+ entries fast, reliably and with minimal tokens, classifies requirements and detects conflicts and duplicates.
- Developing 3D defect classification models under limited data and strong confounding.

Azzulei Technologies — Madrid, Spain
Computer Vision Intern — *Jun – Aug 2025*
- Built a real time multi-object tracking pipeline (YOLOv8, ByteTrack, optical flow) for automated football camera control, robust to a small, fast ball and occlusions.

Imperial College London — Remote (London, UK)
Research Assistant — *Jun – Aug 2024*
- Modelled nonlinear dynamical systems (Lotka–Volterra, SIR, Lorenz) with Neural ODEs and PINNs.

Endesa — Madrid, Spain
Energy Data Analyst Intern — *Jun – Aug 2023*
- Forecasted daily Iberian ancillary service costs using MLP and regression models (MAPE $< 10\%$).

Selected Research

Master's Thesis: Vision-Only Autonomous Drones Trained with Differentiable Simulation — 2026
- Training an FPV drone to avoid obstacles from RGB images alone with SHAC, backpropagating through differentiable JAX physics and a gaussian splat renderer into a CNN + GRU policy.
- Reached 92.5% success on 1,024 evaluation corridors after just 0.9 hours of training on a single GPU.

TopoSIGMA: Spectral Information-Geometric Mapping Architecture — 2025
- Designed an online ROS 2 mapping system combining spectral place discovery, loop closure informed by covariance, DINOv2 descriptors and directional node distributions.
- Manuscript in preparation for submission to Robotics and Autonomous Systems.

Skills

Software: Python (advanced), PyTorch, JAX, ROS 2, Git, Docker, K8S, Azure, LangGraph, n8n
AI & ML: Agentic AI, Computer Vision, Geometric AI, Gaussian Splatting, Differentiable Simulation, NLP, DRL
Mathematics: Linear Algebra, Probabilistic AI, Optimization, Differential Geometry, Statistics
Languages: Spanish (Native), English (Fluent), German (Beginner)

Honors & Awards

Bending Spoons Hackathon Winner — Milan, Italy | 2026
Selected among Spain's top 20 tech students for First Ascent and winner of the event's hackathon.

10th Smart Industry Hackathon Winner — Madrid, Spain | 2025
Built a virtual assistant with recommendations for train operators, in a hackathon featuring Kearney and CAF.

UNIJES Social Tech Challenge Winner — Bilbao, Spain | 2024
Voice-controlled autonomous wheelchair project

Academic Excellence Scholarship — Madrid, Spain | 2021, 2023, 2024
Top-scoring students across Community of Madrid

Baccalaureate Honorific Mention — Madrid, Spain | 2021
2nd highest GPA in Community of Madrid